Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Welcome to the Course

- Welcome to Applied Unsupervised Learning in Python
  Video • 5 min
- MADSwPy Certificate Roadmap
  Reading • 5 min
- Course Syllabus
  Reading • 10 min
- Additional Resources
  Reading • 10 min
- Meet Your Fellow Learners
  Discussion Prompt • 15 min
- Help Us Learn About You
  Reading • 5 min

Lesson 1: Methods for Dimensionality Reduction

- Dimensionality Reduction: A Brief Introduction
  Video • 16 min
- Dimensionality Reduction with Feature Selection: Information Gain
  Video • 15 min
- Time to Practice: Dimensionality Reduction
  Graded Assignment • Grade: 100%
- Dimensionality Reduction with Feature Selection: Principal Component Analysis (PCA) Explained
  Video • 20 min
- Visualizing PCA Results: Foundations
  Video • 8 min
- Visualizing PCA Results: Biplots and Variance Plots
  Video • 17 min
- Ten Quick Tips for Effective Dimensionality Reduction
  Reading • 45 min
- Time to Practice: Principal Component Analysis (PCA)
  Graded Assignment • Grade: 100%
- Singular Value Decomposition (SVD)
  Video • 21 min
- Applications of SVD in Data Science
  Video • 13 min
- Time to Practice: Singular Value Decomposition (SVD)
  Graded Assignment • Grade: 100%
- Manifold Learning: Multidimensional Scaling (Part 1)
  Video • 17 min
- Manifold Learning: Multidimensional Scaling (Part 2)
  Video • 11 min
- Time to Practice: Manifold Learning (Multidimensional scaling: Parts 1 & 2)

Introduction to Module 2 Assignment: Clustering

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For the Module 2 programming assignment, you will focus on clustering algorithms using a dataset of food recipes. Clustering algorithms have many practical applications in real-world scenarios -- from grouping customers together to refine targeted advertisements, to organizing themes in a large set of text documents into topical groups. This week, we're going to be applying clustering algorithms to explore the structure of a dataset of food recipes. We'll begin with k-means clustering, move on to an example of hierarchical clustering, and then finish by exploring the capabilities of DBSCAN.

1. **First,** you will perform k-means clustering on the recipe dataset, evaluating different choices for the number of clusters `k` using the Calinski-Harabasz index to select the optimal value for `k`. Then, you'll find which ingredients are representative for each of the recipe clusters found.
2. **Second,** you are going to work through an example of putting a subset of the recipe data through a hierarchical clustering algorithm and visualize the resulting tree structure.
3. **Third,** you will run DBSCAN to cluster the dataset and explore outliers, using both text and nutritional representations of recipe data.

Go to next item ✓ Completed

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